ETALE LOCALNESS ON THE SOURCE-AND-TARGET IS COMPATIBLE WITH ETALE BASE CHANGE OF THE DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 3 navigation stubs for the named definition and lemmas of the localness notion; the application of each clause is written out inside the span, so nothing load bearing is delegated.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The label lemma-etale-etale-local-source-target is defined only in descent.tex among the staged chapters, so no same-name restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Etale localness on the source-and-target and its equivalent clauses. Same chapter, Definition definition-local-source-target and Lemma lemma-local-source-target-implies, invoked at proof lines 7812-7825 and 7834-7842. Supplies the clauses (1), (3) and (4) used in both directions.
- The locus W(f) attached to a property via the largest open on which the property holds. Same chapter, Lemma lemma-largest-open-of-the-source, invoked at proof lines 7808 and 7812-7813. Supplies the object being compared and the input to which the localness definition is applied.
- An etale morphism is open. Morphisms, the openness of etale morphisms, used at proof lines 7838-7840. Supplies that the image of the etale neighbourhood is open, so that a single-member etale covering is obtained.

Source reference. Original lemma 0CEY: sources/stacks-descent/pinned-original-source.tex, source lines 6890–6890. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04R0: sources/stacks-descent/pinned-original-source.tex, source lines 7399–7399. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $\mathcal{P}$ be a property of morphisms of schemes. We say $\mathcal{P}$ is *étale local on source-and-target* if

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(1) (stable under precomposing with étale maps) if $f : X \to Y$ is étale and $g : Y \to Z$ has $\mathcal{P}$, then $g \circ f$ has $\mathcal{P}$,
(2) (stable under étale base change) if $f : X \to Y$ has $\mathcal{P}$ and $Y' \to Y$ is étale, then the base change $f' : Y' \times_Y X \to Y'$ has $\mathcal{P}$, and
(3) (locality) given a morphism $f : X \to Y$ the following are equivalent
   (a) f has $\mathcal{P}$,
   (b) for every $x \in X$ there exists a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with étale vertical arrows and $u \in U$ with $a(u) = x$ such that h has $\mathcal{P}$.

Lemma 0.2. *Let $\mathcal{P}$ be a property of morphisms of schemes which is étale local on the source-and-target. Given a commutative diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array} \quad \textit{with points} \quad \begin{array}{ccc} x' & \longrightarrow & y' \\ \downarrow & & \downarrow \\ x & \longrightarrow & y \end{array}$$

such that g' is étale at x' and g is étale at y', then $x \in W(f) \Leftrightarrow x' \in W(f')$ where $W(-)$ is as in Lemma 0CEY.

Proof. Lemma 0CEY applies since $\mathcal{P}$ is étale local on the source by Lemma 04R0.

Assume $x \in W(f)$. Let $U' \subset X'$ and $V' \subset Y'$ be open neighbourhoods of x' and y' such that $f'(U') \subset V'$, $g'(U') \subset W(f)$ and $g'|_{U'}$ and $g|_{V'}$ are étale. Then $f \circ g'|_{U'} = g \circ f'|_{U'}$ has $\mathcal{P}$ by property (1) of Definition 0.1. Then $f'|_{U'} : U' \to V'$ has property $\mathcal{P}$ by (4) of Lemma 04R0. Then by (3) of Lemma 04R0 we conclude that $f'_{U'} : U' \to Y'$ has $\mathcal{P}$. Hence $U' \subset W(f')$ by definition. Hence $x' \in W(f')$.

Assume $x' \in W(f')$. Let $U' \subset X'$ and $V' \subset Y'$ be open neighbourhoods of x' and y' such that $f'(U') \subset V'$, $U' \subset W(f')$ and $g'|_{U'}$ and $g|_{V'}$ are étale. Then $U' \to Y'$ has $\mathcal{P}$ by definition of $W(f')$. Then $U' \to V'$ has $\mathcal{P}$ by (4) of Lemma 04R0. Then $U' \to Y$ has $\mathcal{P}$ by (3) of Lemma 04R0. Let $U \subset X$ be the image of the étale (hence open) morphism $g'|'_U : U' \to X$. Then $\{U' \to U\}$ is an étale covering and we conclude that $U \to Y$ has $\mathcal{P}$ by (1) of Lemma 04R0. Thus $U \subset W(f)$ by definition. Hence $x \in W(f)$. □